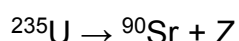


Nuclear medicine uses radiopharmaceuticals for disease treatment and as tracers in medical imaging studies. Yttrium-90 (^{90}Y) is a radiopharmaceutical agent used to treat overgrown joint lining known as pigmented villonodular synovitis, as well as some forms of liver cancer. It has a half-life of about 64 hours.

Although the relatively short half-life of ^{90}Y is optimal for use in medical applications, transportation and storage of the radiopharmaceutical are not feasible. As a result, strontium-90 (^{90}Sr), with a half-life of 28.8 years, is the most common source of ^{90}Y . ^{90}Sr is created by a nuclear fission process that begins with uranium-235 (^{235}U) in a nuclear reactor as shown in Reaction 1.



Reaction 1

Fission of ^{235}U produces ^{90}Sr as well as additional fission products (Z) including, but not limited to, technetium-99 (^{99}Tc), iodine-129 (^{129}I), and zirconium-93 (^{93}Zr). Figure 1 shows the distribution of fission products.

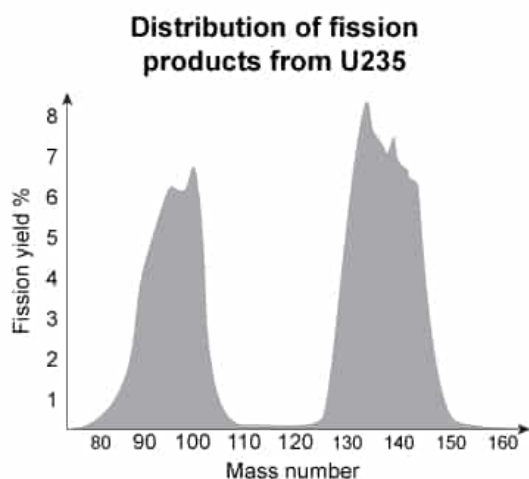


Figure 1 Distribution of ^{235}U fission products by mass number

To create the final ^{90}Y required for nuclear medicine studies, ^{90}Sr decay is carried out in a controlled ^{90}Y generator. ^{90}Y is then separated from residual ^{90}Sr for use in clinical applications.

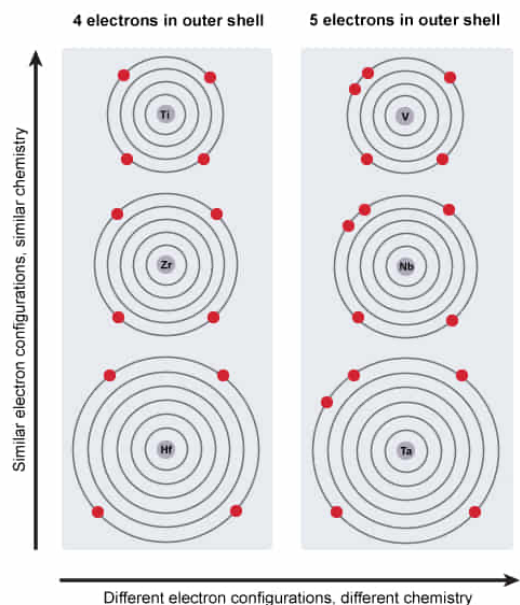
In its nuclear decay chain, ^{90}Sr decays into ^{90}Y , and ^{90}Y then decays to ^{90}Zr . Which of the following elements exhibits chemical properties most similar to zirconium-90?

- ☐ A. ^{90}Nb [30%]
- ✓ ☒ B. ^{174}Hf [63%]
- ☐ C. ^{46}Sc [3%]
- ☐ D. ^{48}Ca [2%]

Correct

63% answered correctly

Explanation:



Note: image depicts only relevant valence electrons

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Atoms are most stable when they have a full outer orbital shell (**valence shell**). For this reason, the chemical behavior of an atom is determined largely by the **number of valence electrons** because this number determines how many electrons must be **gained or lost** to achieve stability. Accordingly, elements in the **same column** of the **periodic table** have the **same valence electron configuration**, and therefore have **similar chemical properties** (ie, react with other elements in similar ways).

In contrast, *physical properties* of elements in the same column are still often dissimilar because physical properties (eg, color, density, atomic radius) are determined by atomic characteristics other than the valence electrons.

^{90}Y decays by β^- decay, which causes the atomic number to increase by 1 and produces zirconium-90 (^{90}Zr). Zirconium is in the same column as hafnium (Hf), and therefore any isotope of hafnium will have similar chemical properties to any isotope of zirconium.

(Choice A) Niobium-90 (^{90}Nb) is in the same row as ^{90}Zr and has the same atomic mass. However, elements in the same row of the periodic table have different numbers of valence electrons and therefore different chemical properties.

(Choice C) Scandium-46 (^{46}Sc) has chemical characteristics most similar to ^{90}Y , not ^{90}Zr .

(Choice D) Calcium-48 (^{48}Ca) has chemical characteristics most similar to ^{90}Sr , not ^{90}Zr .

Educational objective:

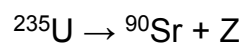
Chemical behavior is governed largely by the number of electrons in an atom's valence shell. Atoms with the same number of valence electrons (ie, in same column of the periodic table) have similar chemical properties.

Time spent:0 sec
Subject:
General Chemistry

QID:400195
System:
Atomic Theory and Chemical Composition

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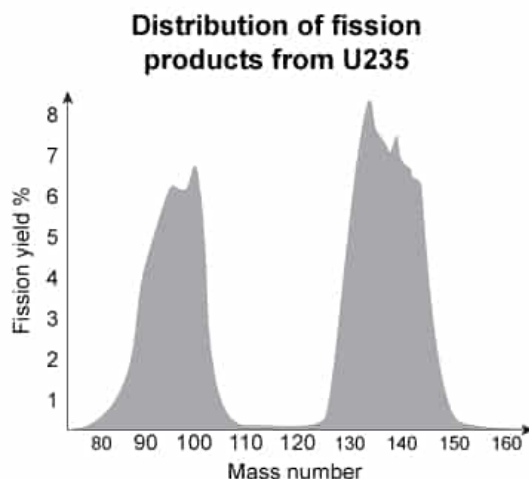
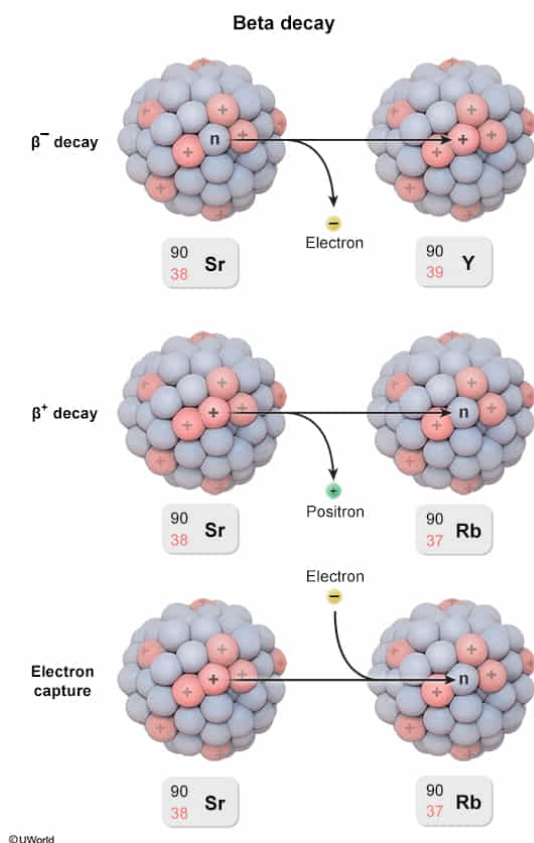


Figure 1 Distribution of ^{235}U fission products by mass number

To create the final ^{90}Y required for nuclear medicine studies, ^{90}Sr decay is carried out in a controlled ^{90}Y generator. ^{90}Y is then separated from residual ^{90}Sr for use in clinical applications.

^{90}Sr is converted to ^{90}Y by which of the following processes?

- ☐ A. Electron capture [9%]
- ☐ B. Positron emission [11%]
- ☐ C. Gamma ray emission [15%]
- ☒ D. Beta minus decay [63%]

Explanation:

The mass of an atom is predominantly contained in the nucleus, where each proton and neutron has an atomic mass of approximately 1 **amu**. Because protons and neutrons have approximately the same mass, converting one to the other does not significantly alter the mass of an atom (ie, the **mass number** does not change). However, such a change does alter the identity of the element, which is determined only by the number of protons in the nucleus (ie, the **atomic number**).

Protons and neutrons can be interconverted during radioactive decay by the following mechanisms:

- **Beta emission** is the ejection of an electron or a positron from the nucleus. The most common form of beta decay is emission of a nuclear electron (ie, β^- decay), in which a neutron converts into a proton and increases the atomic number by one. Less commonly, **positron emission** (ie, β^+ decay) converts a proton into a neutron, decreasing the atomic number by one.
- **Electron capture** is a special form of beta decay in which a nucleus absorbs an electron from an external source without emitting a particle. The absorbed electron converts a proton into a neutron, decreasing the atomic number by one.

The **periodic table** shows that strontium has an atomic number of 38 whereas yttrium has an atomic number of 39. Because the atomic number increases by one, strontium must gain a proton by emitting a nuclear electron (ie, beta emission).

(Choices A and B) Positron emission and electron capture both result in a *decrease* in the

atomic number, which would convert strontium to rubidium.

(Choice C) The **emission of a gamma ray** (ie, a high-energy photon) does not affect the atomic number or mass of the nucleus, but it does decrease its energy.

Educational objective:

Beta decay typically involves the emission of a nuclear electron or positron, but sometimes an electron capture can occur. Electron emission increases the atomic number by one whereas positron emission and electron capture both decrease the atomic number by one.

Time spent:0 sec

Subject:
General Chemistry

QID:400196

System:
Atomic Theory and Chemical Composition